Ch 8: Introduction to trigonomentry

Question Bank | 2M | 3M | 5M

<u>SECTION A - 2 MARKS</u>

1. In triangle ABC right-angled at B, AB = 24 cm and BC = 7 cm. Find sin A, cos A, and tan A. ***(PYQ 2019)***

Q2. In triangle ABC right-angled at B, AB = 5 cm and AC = 13 cm. Find all six trigonometric ratios of angle A.

Q3. In triangle PQR right-angled at Q, PQ = 3 cm and PR = 6 cm. Find sin P, cos P, and tan R.

Q4. If tan A = 4/3, find sin A and cos A where A is an acute angle. ***(PYQ 2019)***

Q5. If sin A = 3/4, find cos A, tan A, and cot A.

Q6. If cos θ = 7/25, find sin θ, tan θ, and sec θ.

Q7. If cot θ = 7/8, evaluate (1 + sin θ)(1 − sin θ) / (1 + cos θ)(1 − cos θ). ***(PYQ NCERT)***

Q8. If 3 cot A = 4, find the value of (4 sin A − 3 cos A) / (2 sin A + 6 cos A).

Q9. If $4 \tan \theta = 3$, find the value of $(4 \sin \theta - \cos \theta + 1) / (4 \sin \theta + \cos \theta - 1)$. ***(PYQ 2018)***

Q10. If $\tan A + \cot A = 6$, find $\tan^2 A + \cot^2 A - 4$. ***(PYQ 2024)***

Q11. Find the value of $2 \tan^2 45° + \cos^2 30° - \sin^2 60°$.

Q12. Find the value of $\sin 60° \cos 30° + \sin 30° \cos 60°$. ***(PYQ 2020)***

Q13. Evaluate: $(\sin 30° + \tan 45° - \operatorname{cosec} 60°) / (\sec 30° + \cos 60° + \cot 45°)$. ***(PYQ 2018)***

Q14. Evaluate: $\sin^2 30° + \cos^2 45° + 4 \tan^2 30° + \sin^2 90° - 2 \cos^2 90°$.

Q15. Evaluate: $\cos^2 45° + \sin^2 45° + \tan^2 60° - \operatorname{cosec}^2 30°$.

Q16. Evaluate: $2 \cos^2 60° + 3 \sin^2 45° - 3 \sin^2 30° + 2 \cos^2 90°$. ***(PYQ)***

Q17. Evaluate: $(\cos 0° + \sin 45° + \sin 30°)(\sin 90° - \cos 45° + \cos 60°)$. ***(PYQ)***

Q18. If $\tan 2A = \cot(A + 18°)$ where $2A$ is an acute angle, find A. ***(PYQ 2018)***

Q19. If $\sin 3A = \cos(A - 26°)$ where $3A$ is an acute angle, find A. ***(PYQ 2020)***

Q20. If $\sec 4A = \operatorname{cosec}(A - 20°)$ where $4A$ is an acute angle, find A. ***(PYQ 2020)***

Q21. If $\tan A = \cot B$, prove that $A + B = 90°$. ***(PYQ 2019)***

Q22. If $\sin(A + B) = 1$ and $\cos(A - B) = 1$ where $0° < A + B \leq 90°$ and $A > B$, find A and B.

Q23. If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = 1/\sqrt{3}$ where $A + B \leq 90°$ and $A > B$, find A and B. *(PYQ 2019)*

Q24. If $\sin(A - B) = 1/2$ and $\cos(A + B) = 1/2$ where $0° < A + B \leq 90°$ and $A > B$, find A and B. *(PYQ 2019)*

Q25. If $\cos(A + B) = 0$ and $\sin(A - B) = \sqrt{3}/2$ where A, B are acute angles, find A and B. *(PYQ 2018)*

Q26. If $\alpha + \beta = 90°$ and $\alpha = 2\beta$, find $\cos^2\alpha + \sin^2\beta$. *(PYQ 2024)*

Q27. If $\tan x - \cot y = 0$, find the value of $x + y$. *(PYQ 2024)*

Q28. If A, B, and C are interior angles of triangle ABC, show that $\sin(B + C)/2 = \cos A/2$. *(PYQ 2019)*

Q29. In a right triangle ABC right-angled at B, if $\tan A = 1$, verify that $2 \sin A \cos A = 1$. *(PYQ NCERT)*

Q30. Find the value of x if $\cos 2x = \sin(x - 30°)$ and 2x is an acute angle. *(PYQ)*

Q31. Find the value of: $(\tan 45° / \operatorname{cosec} 30°) + (\sec 60° / \cot 45°) - (5 \sin 90° / 2 \cos 0°)$. *(PYQ*

<u>SECTION B - 3 MARKS</u>

Q1. Prove: $(\text{cosec}\,\theta - \cot\theta)^2 = (1 - \cos\theta) / (1 + \cos\theta)$. ***(PYQ NCERT)***

Q2. Prove: $\cos A/(1 + \sin A) + (1 + \sin A)/\cos A = 2\sec A$. ***(PYQ 2019)***

Q3. Prove: $\sin\theta/(1 - \cos\theta) + \tan\theta/(1 + \cos\theta) = \sec\theta\,\text{cosec}\,\theta + \cot\theta$. ***(PYQ 2020)***

Q4. Prove: $(\tan\theta + \sec\theta - 1)/(\tan\theta - \sec\theta + 1) = (1 + \sin\theta)/\cos\theta$. ***(PYQ 2019)***

Q5. Prove: $[\tan\theta/(1 - \cot\theta)] + [\cot\theta/(1 - \tan\theta)] = 1 + \sec\theta\,\text{cosec}\,\theta$. ***(PYQ 2019 — most repeated)***

Q6. Prove: $(\sin A + \text{cosec}\,A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$. ***(PYQ 2018)***

Q7. Prove: $(\sin A - 2\sin^3 A)/(2\cos^3 A - \cos A) = \tan A$. ***(PYQ 2019)***

Q8. Prove: $(1 + \cot A - \text{cosec}\,A)(1 + \tan A + \sec A) = 2$. ***(PYQ 2019)***

Q9. Prove: $(\text{cosec}\,A - \sin A)(\sec A - \cos A) = 1/(\tan A + \cot A)$. ***(PYQ 2018)***

Q10. Prove: $\sqrt{[(1 + \sin A)/(1 - \sin A)]} = \sec A + \tan A$. ***(PYQ 2020)***

Q11. Prove: $(\sin\theta + \cos\theta)(\tan\theta + \cot\theta) = \sec\theta + \operatorname{cosec}\theta$. ***(PYQ)***

Q12. Prove: $(1 + \tan^2\theta)(1 - \sin\theta)(1 + \sin\theta) = 1$. ***(PYQ 2020)***

Q13. Prove: $\sin\theta/(\cot\theta + \operatorname{cosec}\theta) = 2 + \sin\theta/(\cot\theta - \operatorname{cosec}\theta)$. ***(PYQ)***

Q14. Prove: $(\tan A + \sin A)/(\tan A - \sin A) = (\sec A + 1)/(\sec A - 1)$. ***(PYQ 2019)***

Q15. If $\operatorname{cosec}\theta + \cot\theta = p$, show that $\cos\theta = (p^2 - 1)/(p^2 + 1)$. ***(PYQ 2019)***

Q16. If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, show that $(m^2 - n^2)^2 = 16mn$. ***(PYQ 2020)***

Q17. If $\sin\theta + \cos\theta = p$ and $\sec\theta + \operatorname{cosec}\theta = q$, show that $q(p^2 - 1) = 2p$. ***(PYQ 2019)***

Q18. If $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$, show that $\cos\theta + \sin\theta = \sqrt{2}\cos\theta$. ***(PYQ 2020)***

Q19. If $a\sec\theta + b\tan\theta = m$ and $b\sec\theta + a\tan\theta = n$, prove that $a^2 - b^2 = m^2 - n^2$. ***(PYQ 2024)***

Q20. If $\cos A + \cos^2 A = 1$, prove that $\sin^2 A + \sin^4 A = 1$. ***(PYQ)***

Q21. If $\sin\theta + \cos\theta = \sqrt{3}$, prove that $\tan\theta + \cot\theta = 1$. ***(PYQ 2020)***

Q22. Prove: $(\sec A - 1)/(\sec A + 1) = [(\sin A)/(1 + \cos A)]^2$. ***(PYQ 2020)***

Q23. Evaluate: $[(\sin 30° + \cos 60°)(\sin 60° + \cos 30°)] / (\sec^2 45° - \cot^2 45°)$. *(PYQ)*

Q24. Evaluate: $[\sin 90° - \cos 45° + 2 \tan 30°] / [\sec^2 30° - \cot^2 60°]$. *(PYQ)*

Q25. Prove: $\tan^2 A - \tan^2 B = (\sin^2 A - \sin^2 B)/(\cos^2 A \cos^2 B)$. *(PYQ HOTS)*

Q26. Prove: $(1 + \sin \theta - \cos \theta)/(1 + \sin \theta + \cos \theta) = \tan(\theta/2)$. *(PYQ HOTS)*

Q27. If $\sin B = \sin Q$ and B and Q are acute angles, prove that $B = Q$. *(PYQ NCERT)*

Q28. In triangle ABC right-angled at C, if $\tan A = 1/\sqrt{3}$, find the value of $\sin A \cos B + \cos A \sin B$. Also verify that $\sin(A + B) = 1$. *(PYQ)*

Q29. Prove: $\sin^2 A + \cos^2 A = 1$. Also use it to prove $\sec^2 A - \tan^2 A = 1$. *(PYQ NCERT)*

Q30. Find x: $2 \sin^2(x + 45°) - \sin^2(30°) - \cos^2(x - 45°) = 0$. *(PYQ HOTS)*

SECTION C - 5 MARKS

Q1. Prove: $[\tan\theta/(1 - \cot\theta)] + [\cot\theta/(1 - \tan\theta)] = 1 + \sec\theta \operatorname{cosec}\theta$. ***(PYQ 2019 — most repeated)***

Q2. Prove: $(\cos A - \sin A + 1)/(\cos A + \sin A - 1) = \operatorname{cosec} A + \cot A$, using the identity $\operatorname{cosec}^2 A = 1 + \cot^2 A$. ***(PYQ NCERT — most repeated)***

Q3. Prove: $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$. ***(PYQ 2018)***

Q4. Prove: $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$. ***(PYQ 2019)***

Q5. Prove: $(\tan\theta + \sec\theta - 1)/(\tan\theta - \sec\theta + 1) = (1 + \sin\theta)/\cos\theta$. ***(PYQ 2019)***

Q6. If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, show that $m^2 - n^2 = 4\sqrt{(mn)}$. Also show $(m^2 - n^2)^2 = 16mn$. ***(PYQ 2020)***

Q7. If $\sin\theta + \cos\theta = p$ and $\sec\theta + \operatorname{cosec}\theta = q$, show that $q(p^2 - 1) = 2p$. Hence find p if $q = 2$. ***(PYQ 2019)***

Q8. Prove: $\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta\cos^2\theta$. ***(PYQ HOTS)***

Q9. Prove: $(\sec\theta + \tan\theta - 1)/(\tan\theta - \sec\theta + 1) = \cos\theta/(1 - \sin\theta)$. ***(PYQ)***

Q10. Prove: $(\sin\theta - \cos\theta + 1)/(\sin\theta + \cos\theta - 1) = 1/(\sec\theta - \tan\theta)$. ***(PYQ NCERT)***

Q11. If cosec θ + cot θ = p, prove cos $\theta = (p^2 - 1)/(p^2 + 1)$. Hence find θ if p = $\sqrt{3}$. ***(PYQ 2019)***

Q12. Prove: $(\sec^2\theta - \sin^2\theta)/(\tan^2\theta + \cos^2\theta) = 1/(\sin^2\theta + \cos^2\theta)$. ***(PYQ)***

Q13. Evaluate: $[4(\sin^4 30° + \cos^4 60°) - 3(\cos^2 45° - \sin^2 90°)]$ / $[2(\sin^2 60° + \cos^2 45°) - \tan^2 30°]$. ***(PYQ 2019)***

Q14. Evaluate: $[2 \sin^2 30° \cos^2 45° + 4 \tan^2 60° + \sin^2 90° + \cos^2 0°]$ / $[\cos^2 60° + \sin^2 45°]$. ***(PYQ 2018)***

Q15. If a cos θ − b sin θ = c, prove that a sin θ + b cos θ = $\pm\sqrt{(a^2 + b^2 - c^2)}$. ***(PYQ HOTS)***

Q16. Prove: (tan A + sin A)/(tan A − sin A) = (sec A + 1)/(sec A − 1). Hence evaluate when A = 45°. ***(PYQ 2019)***

Q17. If x = a sec θ + b tan θ and y = a tan θ + b sec θ, prove that $x^2 - y^2 = a^2 - b^2$. ***(PYQ 2020)***

Q18. If cos θ + sin θ = $\sqrt{2}$ cos θ, show that cos θ − sin θ = $\sqrt{2}$ sin θ. Hence prove that $\cos^4\theta - \sin^4\theta = 2\cos^2\theta - 1$. ***(PYQ 2020)***

Q19. Prove: cosec A/(cosec A − 1) + cosec A/(cosec A + 1) = 2 $\sec^2$A. ***(PYQ)***

Q20. Prove: $(1 - \sin\theta + \cos\theta)^2 = 2(1 + \cos\theta)(1 - \sin\theta)$. ***(PYQ HOTS)***

Q21. In triangle ABC right-angled at B, sin A = 1/2. Find all six trigonometric ratios of A and C. Also verify that $\sin^2 A + \cos^2 A$ = 1. ***(PYQ NCERT)***

Q22. Prove: $(\tan A - \tan B)^2 + (1 + \tan A \tan B)^2 = (\sec^2 A)(\sec^2 B)$. ***(PYQ HOTS)***

Q23. Evaluate step by step: $[2 \tan^2 45° + 3 \operatorname{cosec}^2 30° - \sec^2 60°] / [\sin^2 30° + 4 \cot^2 45° - \sec^2 60°]$. ***(PYQ)***

Q24. Prove: $\cot \theta/(\cot \theta - \cot 3\theta) + \tan \theta/(\tan \theta - \tan 3\theta) = 1$. ***(PYQ HOTS)***

Q25. Prove: $[(\sin \theta + \cos \theta)(1 - \sin \theta \cos \theta)] = \sin^3\theta + \cos^3\theta$. Also verify for $\theta = 30°$. ***(PYQ)***